Working Principle: The sunlight incident on the glass plate passes through and reaches the junction. An incident light photon at the junction may collide with a valence electron and impart sufficient energy to make a transition to the conduction band. As a result, an electron-hole pair is formed. The newly formed electrons are minority carriers in the *P* region. They move freely across the junction. Similarly, holes formed in the *N-region* cross the junction in the opposite direction. The flow of these electrons and holes across the junction is in a direction opposite to the conventional forward current in a *PN* junction. Further, it leads to the accumulation of majority carriers on both sides of the junction.

In bright sunlight, 0.6 V is developed by a single solar cell. The amount of power the cell can deliver depends on the extent of its active surface. An average cell will produce about 30 mW per square inch of surface, operating in a load of 4 Ω. To increase the power output, large banks of cells are used in series and parallel combinations.

Application

1. It is used to provide electrical power to satellites and space vehicles
2. It is used to energize these storage cells
3. It is also used in home for electrical requirements.

6.8. UNIPOLAR JUNCTION TRANSISTOR (UJT)

UJT is a three terminal semiconductor switching device. As the name indicates, it has only one *PN* junction (i.e. unijunction). UJT can be used to control a large AC power with small gain, and not to be used as amplifier. It exhibits negative resistance characteristics and can be used as an oscillator.

History: The UJT was invented as a by-product of research on germanium tetrode transistors at General Electrical Company. It was patented in 1953.

Construction: Figure 6.8 shows the basic structure of UJT and its symbol. It consists of lightly doped N-type silicon bar with ohmic contacts at the two ends, called base1 (B_1) and base2 (B_2). A P-type emitter is diffused nearer to the base2 of the bar, and this forms a *PN* junction diode with the base as shown in Figure 6.8(a). The *P-type* region is called emitter (*E*), since the device contains two base terminals with one *PN* junction and hence it is also called as double base diode. The resistance (R_{BB}) between the

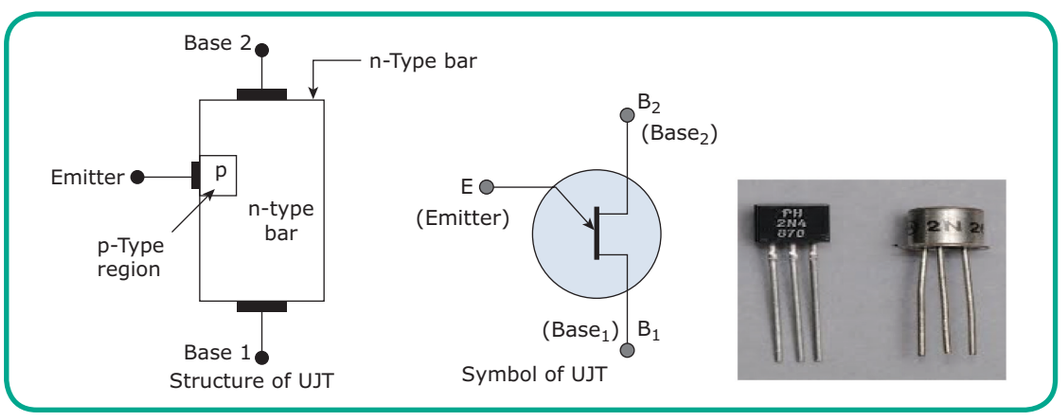


Figure 6.8(a) **Basic Structure of Unijunction Transistor and its Symbol**

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